HAZOP Study Manager – Quick Guide

Offline relational HAZOP database · part of the Process Safety Analysis Toolkit

1 · What it does

The HAZOP Study Manager records a complete HAZOP study as related tables: study details and team, sessions, process safety information (PSI), nodes, deviations (one risk assessment per deviation), causes, a safeguard register with per-deviation IPL claims, and a recommendation register. Deleting a node removes its deviations and links, so no orphan records remain. It includes the guideword library for the chemical, steel and metals industries (57 parameters, 92 parameter / guideword combinations, 629 potential causes).

2 · Workflow

1. **Starting from a What-If?** – open the HAZOP study file exported by the What-If / Checklist tool: the nodes and deviations are already seeded; assign parameter and guideword to each.
2. **Study details & team** – title, scope, design intent, facilitator, scribe, team members, session attendance.
3. **Process safety info** – drawings, procedures and data used; link each document to its nodes (one click adds a typical PSI checklist).
4. **Nodes** – number, name, drawings, battery limits, design intent, design and operating parameters, previous incidents and MOC references.
5. **HAZOP worksheet** – choose the node; add deviations one by one or many at once with *Add deviations from guideword set*. Click a cell to select it; double-click or press Enter to edit it; ✎ opens the deviation editor.
6. **Registers** – review safeguards / IPLs, the **LOPA study list**, track recommendations to close-out, and answer the **parking lot** questions.
7. **Reports & export** – printable report (PDF) and Excel export.

3 · The deviation editor

Section	How to use it
Deviation	Choose parameter and guideword – the deviation text is built automatically (it can be edited).
Causes	Tick causes from the library for that parameter / guideword, or type your own. Several causes per deviation.
Consequences	Describe the worst credible outcome without safeguards.
Risk before safeguards	Rate the severity for each **receptor** – safety, environment, asset, reputation, community. The overall severity is the highest one (the worksheet shows which receptor drives it, e.g. 5^{S}). Then choose the likelihood. The risk is ranked on the 5 × 5 matrix (editable on the *Risk matrix* screen, with a descriptor for each receptor and level).

Safeguards	Type or pick a safeguard – new ones are added to the register and appear in the pull-down list for every other deviation. Set the drop-down to **Safeguard** or **IPL** (an independent protection layer for this scenario: specific, independent, dependable, auditable). For an IPL, tick **LOPA** if it must be checked in a LOPA – the scenario then appears on the *LOPA study list*.
Risk after safeguards	Residual severity by receptor × likelihood. *Copy severities* repeats the ratings from the stage before.
Recommendations	Create a new recommendation or link an existing one (one recommendation can close several deviations).
Risk after recommendations	The risk expected once the recommendations are in place – shows whether they bring the risk down far enough.
Parking lot	A question the team cannot answer at the table: write it down with an owner and move on. Close it on the *Parking lot* screen when it is answered.

> Safeguards can also be entered directly in the worksheet row: type or pick in the box under the Safeguards column and press Enter; use the Safeguard / IPL drop-down and the LOPA tick on each line. The LOPA column of the worksheet flags scenarios sent to the LOPA study list.

4 · Worksheet keys, columns and filters

Key	What it does
Arrows, Tab	Move between cells
Enter or F2	Edit the cell (Shift+Enter for a new line; Esc cancels). Causes, safeguards and recommendations open the deviation editor.
1 – 5	Set S or L directly (S rated by receptor is changed in the editor)
Delete	Clear the cell
Ctrl+C / Ctrl+V	Copy and paste one cell
Ctrl+Enter	Add a deviation below and type its parameter
Alt+↑ / Alt+↓	Move the row

Columns… shows, hides or renames worksheet columns to match the client's template (saved with the study). The filter box and the *Show* list narrow the rows, e.g. High / Very High after safeguards with no recommendation, or rows with open parking-lot items.

5 · LOPA study list

Each scenario with at least one IPL ticked **LOPA** becomes a LOPA scenario (L-01, L-02 …): initiating causes → consequence (the top event), with the IPLs to credit. Record the LOPA status and reference there. The list is also a sheet in the Excel export and in the HAZOP Study Workbook. The LOPA itself is a separate study.

6 · Saving, workbook and import

- Work is kept automatically in the browser of the computer in use.
- To keep the study on the USB stick: **File ▸ Save study as…** and choose the `My studies` folder – every change is then written to that `.hazop.json` file (Chrome / Edge).
- On another computer: **File ▸ Open study file…**. Use **Download backup** regularly as a second copy.
- Teams that record in Excel use the **HAZOP Study Workbook** (.xlsx, or .xlsm with macros). **File ▸ Import HAZOP workbook** reads a filled-in workbook back into the tool – study, team, PSI, nodes, deviations, causes, safeguards with IPL and LOPA flags, risk and recommendations. Deviation wording is rebuilt from parameter and guideword.

7 · Example

File ▸ Load example study loads a hot DRI transfer study: node N01 the hot DRI transfer conveyor (shaft discharge to EAF charging bin) and node N02 the N2 fire detection, inerting & extinguishing system, with team, PSI, seven deviations rated by receptor before safeguards, after safeguards and after recommendations, a shared safeguard register with IPL claims, seven recommendations and three parking-lot questions. It is the same scenario as the fault tree, event tree and bow tie examples. The study is also in `Examples/Example – Hot DRI transfer conveyor.hazop.json`. Data are illustrative.

8 · Links to the other tools

The **What-If / Checklist Analysis** before the HAZOP sends its questions to HAZOP nodes (export to a HAZOP study file). A high-risk HAZOP scenario whose protection relies on an instrumented or engineered system can be quantified in the **Fault Tree & PFD Calculator** (PFDavg of the protection), its outcomes in **Event Tree Analysis**, and the resulting barriers managed in **Bow Tie Analysis**.